Interest Rate Shock Transmission Across Yield Curves and Asset Markets: Evidence from India and the United States (2009–2019)

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Abstract

This paper studies how structural shocks to different segments of the yield curve propagate through financial markets and affect exchange rates and equity returns. We estimate mean-reverting dynamics for three yield segments (short, medium, long), extract conditional volatility via GARCH(1,1), and identify structural shocks with a block-recursive contemporaneous matrix augmented by sign restrictions. The main analysis focuses on India (2009–2019). The United States is used as a robustness check. Identification avoids Cholesky recursion and uses economically motivated zero and sign restrictions; inference is based on residual bootstrap. Results show that short-term dynamics and FX shocks play a prominent role in India, while the U.S. system is comparatively insulated.

1 Motivation

Interest rates at different maturities play a central role in how financial markets process macroeconomic information. In an emerging economy such as India, short-, medium-, and long-term government yields reflect distinct economic forces, ranging from liquidity conditions and policy expectations to long-horizon risk premia. A further motivation for this study is conceptual: term-structure models such as Vasicek and its extensions describe interest rates as mean-reverting processes with different adjustment speeds across maturities, generating a cascading pattern whereby short-term rates respond most immediately to shocks, medium-term rates adjust more gradually, and long-term yields move only sluggishly. Understanding whether such a cascading effect appears in actual data is a key starting point for studying how the yield curve interacts with broader asset markets.

This project addresses this gap by developing an internally identified framework to extract structural interest-rate shocks and analyse how they propagate across the Indian term structure and into FX and equity markets. The approach follows the logic of the advanced empirical macro lectures: when external shocks are unavailable, identification must be achieved within a Structural VAR using economic structure alone. Specifically, the maturity hierarchy provides a natural ordering of contemporaneous influences, enabling a block-recursive A_0 matrix, while theory-consistent

sign restrictions ensure that yield-curve movements satisfy monotonicity on impact. This avoids reliance on Cholesky identification, which imposes arbitrary contemporaneous causality across yields, exchange rates, and equities.

A second motivation is to characterise the internal dynamics of the yield curve itself. Using AR(1) mean-reversion models, HP-filter trends, and GARCH-based conditional volatility, the project quantifies differences in persistence and uncertainty across short-, medium-, and long-term Indian rates. This complements the structural analysis by providing insight into the underlying processes that generate yield movements.

Finally, the United States term structure is estimated in parallel as a robustness exercise. The same pipeline—mean reversion, volatility modelling, VAR estimation, and structural identification—is applied to US data to verify that the identification scheme and dynamic responses behave consistently across a mature financial market. This comparison strengthens the credibility of the Indian results while maintaining India as the primary object of analysis.

2 Literature Review

The empirical term-structure literature highlights that yields at different maturities adjust at distinct speeds and are influenced by different economic forces. Classical affine models such as Vasicek (1977) and Cox, Ingersoll, and Ross (1985) formalise mean reversion in short rates, while modern empirical work (Diebold and Li, 2006) shows that persistence differs across maturities, motivating reduced-form AR(1) representations such as those used here.

The methodological foundation of this project draws on the structural VAR literature. Sims (1980) introduced VARs as flexible systems for macro analysis, and subsequent work developed identification through short-run, long-run, and sign restrictions. Sign-restricted approaches (Uhlig, 2005; Rubio-Ramírez et al., 2010) allow identification without imposing arbitrary recursive orderings, making them particularly suitable when no external shock series exists. The block-recursive structure used here mirrors this framework by restricting contemporaneous interactions only within the yield block while letting FX and equity respond contemporaneously to yield-curve shocks.

Finally, cross-asset spillover studies emphasise the importance of interest-rate innovations for currency and equity pricing. Interest differentials influence exchange-rate dynamics through risk-premium channels (Lustig, Roussanov, and Verdelhan, 2011), while discount-rate shocks affect expected equity returns (Campbell and Shiller, 1988). By estimating maturity-specific structural shocks and tracing their propagation into INR/USD and NIFTY returns, this study contributes to a growing literature connecting term-structure movements with broader asset-market fluctuations in emerging economies.

3 Methodology

3.1 Data Construction and Sample Period

The empirical analysis uses weekly Indian and United States financial data from January 2009 to December 2019 to avoid GFC and Covid Shocks. Daily datasets for interest rates, the USD–INR exchange rate, and equity indices are converted to weekly frequency by selecting the final observation of each week. All datasets are imported into MATLAB, converted into timetables, and synchronized on a common weekly calendar. Incomplete weeks are removed, and isolated missing values are interpolated. Yield levels are retained for mean-reversion estimation, while weekly yield changes, foreign exchange log returns, and equity log returns are used for structural identification and dynamic analysis.

The final dataset includes the 3-month, 2-year, and 10-year yields; FX and equity returns; HP-filtered yield gaps; and conditional volatilities.

3.2 Variable Definitions and Economic Interpretation

The dataset contains the following weekly variables:

Short-term interest rate (r_t^S). This is the 3-month government security yield for India (or equivalent short-term Treasury yield for the United States). This maturity reflects the very front end of the term structure and is most sensitive to monetary policy, liquidity conditions, and short-horizon expectations.

Medium-term interest rate (r_t^M). This is the 2-year government bond yield. It reflects expectations about short-term policy rates over multiple quarters and incorporates information about the slope of the yield curve.

Long-term interest rate (r_t^L). This is the 10-year government bond yield. It captures long-horizon inflation expectations, growth fundamentals, and term premium.

Foreign exchange return (fx_t). We use the weekly log return of the USD–INR exchange rate, expressed such that a positive value corresponds to a depreciation of the rupee.

Equity return (eq_t). We use the weekly log return on the NIFTY 50 index, and on the S&P 500 for the U.S. robustness analysis.

Yield gaps. For each maturity $i \in \{S, M, L\}$,

$$gap_{i,t} = r_{i,t} - \tau_{i,t},$$

where $\tau_{i,t}$ is the smooth trend extracted using the **Hodrick–Prescott filter**.

Conditional volatility. From AR(1)–GARCH(1,1) models, each yield series generates a conditional standard deviation $\sigma_{i,t}$, representing time-varying interest-rate uncertainty at the short, medium, and long ends of the curve.

These variables collectively provide a high-frequency representation of the joint dynamics of the yield curve, foreign exchange market, and equity market.

3.3 Mean Reversion and Volatility Modelling

Interest rates at each maturity i follow a mean-reverting autoregressive process,

$$r_{i,t} = \mu_i + \kappa_i(r_{i,t-1} - \mu_i) + \varepsilon_{i,t}, \tag{1}$$

where κ_i is the speed of mean reversion and $\varepsilon_{i,t}$ is the innovation. Smooth long-run components $\tau_{i,t}$ are extracted using the Hodrick–Prescott filter, and deviations are computed as

$$gap_{i,t} = r_{i,t} - \tau_{i,t}. \tag{2}$$

To capture time-varying uncertainty, AR(1) residuals $u_{i,t}$ are modelled using a GARCH(1,1) specification,

$$u_{i,t} = \sigma_{i,t} z_t, \tag{3}$$

$$\sigma^2_{i,t} = \omega_i + \alpha_i u^2_{i,t-1} + \beta_i \sigma^2_{i,t-1}, \tag{4}$$

yielding conditional volatility series for short-, medium-, and long-term yields.

3.4 Structural VAR Framework

To examine the propagation of interest-rate shocks across maturities and their spillovers into foreign exchange and equity markets, a structural vector autoregression is estimated. Let

$$y_t = \begin{bmatrix} r^S_t \\ r^M_t \\ r^L_t \\ fx_t \\ eq_t \end{bmatrix}, \tag{5}$$

where r^S_t, r^M_t, and r^L_t denote the 3-month, 2-year, and 10-year yields respectively.

The reduced-form VAR is given by

$$y_t = A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad u_t \sim N(0, \Sigma_u), \tag{6}$$

with lag length selected using the Bayesian Information Criterion.

3.5 Structural Representation

The structural form is

$$A_0 y_t = B_1 y_{t-1} + \cdots + B_p y_{t-p} + \varepsilon_t, \tag{7}$$

where the orthogonal structural shocks,

$$\varepsilon_t = \begin{bmatrix} \varepsilon_t^S \\ \varepsilon_t^M \\ \varepsilon_t^L \\ \varepsilon_t^{fx} \\ \varepsilon_t^{eq} \end{bmatrix}, \qquad E[\varepsilon_t \varepsilon_t'] = I, \tag{8}$$

represent unanticipated innovations at each segment of the yield curve and in asset markets.

3.6 Identification: Block-Recursive Structure with Sign Restrictions

Identification is achieved by combining a block-recursive zero structure with economically motivated sign restrictions. The contemporaneous impact matrix is specified as

$$A_0 = \begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} & 0 \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} \end{bmatrix}, \tag{9}$$

reflecting the theoretical maturity hierarchy.

To refine identification, sign restrictions are imposed on impact impulse responses. A positive short-rate shock must raise all yields:

$$\varepsilon_t^S > 0 \quad \Rightarrow \quad \Delta r_t^S(0) > 0,\ \Delta r_t^M(0) \geq 0,\ \Delta r_t^L(0) \geq 0. \tag{10}$$

Similarly for medium- and long-term shocks.

Over-Identification

Zero restrictions and sign restrictions together deliberately over-identify the system, producing a distribution of admissible A_0 matrices rather than a single decomposition. Inference is based on the variability across accepted structural draws, consistent with modern sign-restricted SVAR methodology.

3.7 Structural Shocks

Structural shocks are given by

$$\hat{\varepsilon}_t = A_0 \hat{u}_t. \tag{11}$$

3.8 Bootstrapped Inference

Residual bootstrap draws

$$y_t^* = A_1 y_{t-1}^* + \cdots + A_p y_{t-p}^* + u_t^*, \tag{12}$$

where u_t^* are sampled from VAR residuals. Identification is repeated for each bootstrap draw to construct confidence intervals from the empirical distribution of IRFs.

4 Results

4.1 Yield Curve Dynamics and Volatility Patterns in India

Figures A.1–A.3 document the behaviour of the Indian sovereign yield curve across short (3M), medium (2Y), and long (10Y) maturities over 2010–2019. Together, these figures provide descriptive evidence on the level, variation, and conditional volatility of interest rates prior to our multivariate structural analysis.

Yield Curve Levels. Figure A.1 displays the evolution of the 3-month, 2-year, and 10-year yields. The term structure is generally upward sloping, with long rates consistently above short rates, consistent with standard expectations–hypothesis interpretations. A prominent feature is the sharp spike around mid-2013, corresponding to the "taper tantrum" episode, during which Indian short-term yields briefly surged above 11%. This disturbance is visible across all maturities, but disproportionately so for the short end of the curve, indicating strong monetary tightening pressures. Following 2014, all maturities exhibit a persistent downward drift, reflecting the transition toward a lower-inflation, lower-rate macroeconomic regime.

Yield Changes. Figure A.2 plots weekly yield changes for the three maturities. The distribution of changes is centred near zero, with frequent small fluctuations but occasional large movements. Volatility is elevated between 2011–2014, again with the 2013 episode standing out as a period of extreme short-rate repositioning. The magnitude of weekly adjustments is largest for short rates and smallest for long rates, consistent with the notion that the short end of the curve reacts most strongly to immediate macroeconomic and policy news. The post-2015 period shows markedly dampened yield change dynamics, reflecting calmer macro-financial conditions.

Conditional Volatility. Figure A.3 presents GARCH(1,1)-based conditional volatility estimates for the same set of yields. Volatility is highly clustered over time, with clear bursts around 2009, 2011, and an exceptionally pronounced spike in 2013. The short-rate volatility exceeds that of

medium and long maturities throughout, which is consistent with the term structure literature documenting greater sensitivity of short-term rates to policy and liquidity shocks. After 2014, conditional volatility falls to persistently low levels for all maturities, indicating a more stable interest-rate environment.

Overall, Figures A.1–A.3 confirm three key empirical regularities: (i) yields exhibit persistent mean-reverting behaviour with occasional large dislocations; (ii) short rates are more volatile and more responsive than medium and long maturities; and (iii) volatility is strongly time-varying, validating our use of GARCH models and structural VARs with bootstrapped inference in subsequent sections.

4.2 Distribution of Identified Structural Shocks

Figure A.4 reports the empirical probability density functions (PDFs) of the five structural shocks recovered from the recursively constrained structural VAR for India: the short-rate shock, medium-rate shock, long-rate shock, FX return shock, and equity return shock. These histograms provide a diagnostic assessment of the identification procedure and shed light on the underlying economic propagation mechanisms.

Short-Rate Shock. The short-rate shock distribution is approximately symmetric and centred around zero, with slightly heavier right tails. This indicates that while typical monetary or liquidity disturbances are modest, there are occasional episodes of unusually large upward policy pressure—consistent with the volatility spikes seen in Figure A.3. The distribution shows no severe skewness, suggesting that the identified shock behaves like a conventional orthogonalized monetary/short-end innovation.

Medium-Rate Shock. The medium-rate shock exhibits a similar bell-shaped structure but with a mild left skew. This negative skewness suggests that medium-term yields more often experience downward surprises relative to upward surprises. Such a pattern is consistent with expectations-driven adjustments or growth-related news that tend to lower medium maturities more frequently than they raise them.

Long-Rate Shock. Long-rate shocks are tightly concentrated around zero with comparatively thin tails, indicating that unexpected movements at the long end of the curve are both less frequent and smaller in magnitude. This aligns with the well-known empirical regularity that long-term yields move sluggishly and embody smoothing over expectations of future short rates and inflation.

FX Return Shock. The FX shock distribution is wider than the term-structure shocks and shows mild positive skewness. This reflects that exchange-rate innovations can be large and are often asymmetric, with depreciation episodes occurring with greater magnitude than appreciation

events. The heavier tails confirm the importance of allowing for non-Gaussian disturbances in the VAR, validating the bootstrap inference strategy used later.

Equity Return Shock. The equity shock distribution is the most leptokurtic among the five, with clear fat tails and a slightly left-skewed centre. This behaviour is consistent with the stylised features of equity returns globally: frequent small movements and occasional extreme losses. The non-normality observed here underscores the appropriateness of our identification and inference procedures, which do not rely on Gaussianity and instead use heteroskedasticity and recursiveness to uncover structural dynamics.

Overall, the shock distributions in Figure A.4 confirm that the identification strategy produces economically interpretable innovations with realistic asymmetry and tail behaviour. These properties are essential for generating reliable impulse responses and forecast-error variance decompositions in the subsequent analysis.

4.3 Impulse Response Functions: India

Figures A.5–A.9 present the impulse response functions (IRFs) for India, showing the dynamic effects of the five identified structural shocks—equity, FX, long-rate, medium-rate, and short-rate shocks—on short-, medium-, and long-term interest rates, equity returns, and the exchange rate. Across all shocks, the responses are generally transitory, with most effects dissipating within four to six weeks, indicating rapid adjustment in Indian financial markets.

Response to Equity Shock (Figure A.5)

An equity-market shock generates mild and short-lived effects on the yield curve and exchange rate. Short-term rates decline briefly on impact, bottoming out within two weeks before returning to zero. Medium- and long-term yields respond with slightly smaller amplitudes, indicating limited transmission from equity markets to bond pricing. FX returns show a small appreciation following the shock, but confidence bands include zero throughout, suggesting statistical insignificance. Equity returns themselves display a sharp immediate rise, which quickly dissipates. Overall, the equity shock does not propagate strongly across other financial variables.

Response to FX Shock (Figure A.6)

An exchange-rate shock produces modest but identifiable spillovers. Short- and medium-term rates decline slightly following the depreciation, consistent with the easing of monetary conditions. Long-term yields exhibit minimal response and revert rapidly to baseline. FX returns overshoot initially—as expected under an uncovered interest parity adjustment— before stabilising within four weeks. Equity returns exhibit a small negative reaction, indicating limited transmission from currency markets to domestic equity valuations.

Response to Long-Rate Shock (Figure A.7)

A structural long-rate shock generates the clearest cross-market transmission. Short- and medium-term rates increase on impact, reflecting upward repricing of the entire yield curve. These effects gradually fade but remain economically meaningful for several weeks. FX returns display a small depreciation, consistent with capital outflows or higher domestic discount rates. Equity returns show a mild negative reaction, reflecting the sensitivity of equity valuations to long-term interest-rate shocks. Confidence intervals widen rapidly, however, indicating moderate statistical precision.

Response to Medium-Rate Shock (Figure A.8)

Medium-term interest-rate shocks produce immediate upward responses in both short- and long-term yields, though the effects decay more rapidly than in the long-rate shock. FX returns depreciate slightly following the shock, and equity returns fall modestly on impact. These patterns indicate that medium-term yield shocks transmit through the term structure and tighten financial conditions.

Response to Short-Rate Shock (Figure A.9)

Short-rate shocks produce the strongest but most short-lived movements across financial variables. All segments of the yield curve rise sharply on impact—reflecting the anchoring role of the short-term rate—before reverting to baseline within three to four weeks. FX returns depreciate slightly, while equity returns show a temporary negative reaction. The speed of mean reversion highlights the rapid pass-through of monetary policy surprises in India.

In summary, the Indian IRFs demonstrate that structural shocks influence financial variables primarily in the short run. Interest-rate shocks—particularly short- and long-rate shocks— generate the most pronounced and broad-based responses, whereas equity and FX shocks have more limited and short-lived effects. These findings underscore the importance of domestic interest-rate dynamics in shaping short-term financial conditions in India.

4.4 Forecast Error Variance Decomposition: India

Figures A.10–A.14 report the forecast error variance decomposition (FEVD) for the Indian financial variables. The FEVD quantifies the relative contribution of each structural shock—short, medium, long, FX, and equity—to the variability of interest rates and returns over a 20-week horizon.

FEVD of Equity Returns (Figure A.10). Figure A.10 shows that long-rate shocks are the dominant drivers of Indian equity-return volatility, accounting for roughly 15% of the forecast variance at longer horizons. Medium- rate shocks contribute about 13%, closely followed by short-rate shocks (12–13%). FX and equity shocks play a more modest role, each explaining around 10–11%. These patterns imply that equity-market fluctuations in India are most sensitive to term-structure movements, particularly at the long end, consistent with a discount-rate channel.

FEVD of FX Returns (Figure A.11). Figure A.11 reveals that FX returns are driven primarily by long-rate shocks, which explain approximately 15% of forecast variance. Equity shocks and FX shocks themselves each account for around 13–14%. Short- and medium-rate shocks explain smaller shares (approximately 11%). This distribution suggests that the exchange rate reacts primarily to shifts in the Indian yield curve, rather than to domestic equity or foreign-exchange–specific disturbances.

FEVD of the Long-Term Rate (Figure A.12). Figure A.12 indicates that short-rate shocks are the single most important determinant of long-term yield volatility, contributing close to 30% of the total variance. Medium-rate shocks explain about 26%, while FX shocks account for roughly 17%. Equity shocks play a modest role (around 10%), and long-rate shocks themselves explain only around 2% of variance. This pattern suggests that movements at the long end of the Indian yield curve are largely driven by monetary-policy–related short-rate innovations and medium-term rate adjustments.

FEVD of the Medium-Term Rate (Figure A.13). Figure A.13 shows that medium-term yields are most affected by short-rate and FX shocks. Short-rate shocks explain around 29% of variance, while FX shocks account for approximately 25%. Equity shocks contribute about 16%, long-rate shocks around 11%, and the medium-rate shock itself only 2–3%. These results highlight that the medium-term interest rate in India is jointly shaped by domestic monetary conditions and external FX-related disturbances.

FEVD of the Short-Term Rate (Figure A.14). Figure A.14 demonstrates that short-term interest-rate volatility is driven primarily by FX and long-rate shocks, which explain approximately 23% and 20% of the variance, respectively. Equity shocks contribute around 16%, while medium-rate shocks explain roughly 11%. The short-rate shock itself accounts for less than 2% of the variance. This indicates that although the short rate is the policy instrument, its own innovations play a limited role in its volatility, which is instead shaped by external and longer-maturity financial disturbances.

Overall Interpretation. Taken together, Figures A.10–A.14 reveal a consistent structure: (1) long-rate and short-rate shocks are the primary drivers of financial variability in India; (2) FX shocks play a substantial role, especially for short- and medium-term rates; and (3) equity shocks contribute the least across all variables. These findings underscore the importance of interest-rate dynamics and external sector linkages in shaping India's financial-market behavior.

Robustness Check: United States

Figure A.15 reports the GARCH(1,1) yield volatility for U.S. 3M, 2Y, and 10Y government securities. Volatility declines sharply after 2009 across all maturities, stabilizing at historically low

levels during 2012–2014 when the Federal Reserve maintained near-zero policy rates. The 10Y yield exhibits consistently higher volatility, driven by longer duration and sensitivity to macroeconomic expectations, while short-term rates remain tightly anchored. Observable volatility spikes in 2010, 2011, and 2013 correspond to well-known policy uncertainty periods such as the Eurozone crisis, the U.S. debt-ceiling shock, and the "taper tantrum."

Figure A.16 shows the levels of U.S. yields across the term structure. Short-term yields remain at or near the lower bound until 2015, reflecting the Fed's zero-rate policy. Long-term yields display considerably greater variation—rising during the post-crisis recovery, falling during global risk episodes, and compressing steadily in 2018–2019 as recession concerns intensified. These dynamics contrast sharply with India, where yield spikes are concentrated around specific domestic liquidity and policy events.

Figure A.17 presents weekly changes in U.S. yields. Daily variation for the 10Y maturity is noticeably larger than for short-term rates, but extreme jumps are rare compared with India. Yield changes remain centered around zero with stable dispersion, indicating smoother interest-rate adjustment and deeper market liquidity. These patterns confirm that U.S. rate dynamics are less prone to high-frequency shocks, supporting the robustness of the Indian results.

Figure A.18 presents the distribution of the five structural shocks recovered from the U.S. SVAR. All shocks exhibit approximately symmetric, bell-shaped distributions with limited skewness, confirming that the identification strategy produces economically plausible disturbances.

The short-rate shock displays a tight, centered distribution with thin tails, reflecting the stability of U.S. monetary policy during the sample period and the limited frequency of extreme short-term interest rate movements. Medium- and long-rate shocks show slightly wider dispersion, consistent with the greater sensitivity of longer maturities to macroeconomic news, forward guidance, and risk sentiment.

FX return shocks appear more dispersed than interest-rate shocks, capturing exchange-rate adjustments driven by global risk cycles, safe-haven flows, and monetary-policy divergence episodes. Meanwhile, the equity-return shock distribution is relatively symmetric with moderate variance, indicating that the identification recovers shocks that align well with known episodes of U.S. equity market volatility.

Overall, these distributions validate that the structural shocks extracted for the U.S. sample behave as theoretically expected, supporting the robustness and comparability of the results relative to the India baseline.

Robustness Check: Forecast Error Variance Decomposition (US)

The FEVD results for the United States serve as a robustness check on the main findings for India. By quantifying the relative contribution of each structural shock to the forecast variance of interest rates, exchange rates, and equity returns, these results allow us to assess whether the Indian economy exhibits idiosyncratic patterns or whether similar dynamics arise in an advanced economy. Figures A.24–A.28 present the FEVDs for the United States across all variables.

Equity Return FEVD (US) — Figure A.24. Figure A.24 shows that US equity return variance is fairly evenly distributed across shocks. Equity shocks contribute the largest share (approximately 15%), followed by FX shocks (around 14%) and short- and medium-rate shocks (12–13%). Long-rate shocks account for roughly 10%. The relative stability of these shares across horizons indicates that shock transmission in US equity markets is persistent but not dominated by any single macro-financial shock. Compared with India, the US equity market exhibits a more balanced distribution, consistent with a deeper and more diversified financial system.

FX Return FEVD (US) — Figure A.25. As shown in Figure A.25, long-rate shocks account for the largest share of FX return variance (approximately 15%), closely followed by equity shocks (about 13–14%). Medium- and short-rate shocks contribute moderately (11% and 10%), while FX shocks contribute around 13%. This distribution highlights the sensitivity of the USD exchange rate to movements in longer-term Treasury yields — a pattern consistent with global safe-asset demand channels. In contrast to India, the US FX market places greater weight on long-rate shocks and a smaller weight on short-rate shocks, reflecting differences in monetary policy transmission.

Long-Term Yield FEVD (US) — Figure A.26. Figure A.26 demonstrates that long-term US yields are overwhelmingly driven by **short-rate shocks**, which account for more than 32% of the variance. Medium-rate shocks explain an additional 25%. FX and equity shocks each contribute around 10%, while long-rate shocks explain less than 2%. These shares confirm the expectations hypothesis and the dominant role of policy-rate expectations in shaping the US yield curve. Compared with India, the US long-end of the curve is more tightly anchored to short-term monetary shocks, consistent with a more credible monetary policy regime.

Medium-Term Yield FEVD (US) — Figure A.27. In Figure A.27, the variance of medium-term yields is dominated by short-rate shocks (around 48%). FX and equity shocks contribute around 12–13% each, medium shocks account for roughly 3%, and long shocks explain around 5%. These results reinforce the centrality of the short-term policy stance in shaping the entire US yield curve. Relative to India, the US medium-term segment shows far greater sensitivity to short-rate innovations and far less sensitivity to long-rate shocks.

Short-Term Yield FEVD (US) — Figure A.28. Figure A.28 indicates that short-rate variance in the US is mainly driven by **long-rate shocks** (around 28%) and equity shocks (around 18%), followed by FX shocks (14%) and medium-rate shocks (10%). Short-rate shocks themselves contribute only about 1%. This pattern differs sharply from India, where short-rate movements reflect domestic money market dynamics. In the US, the short-term rate is closely tied to market expectations about long-term economic conditions and global financial factors, consistent with the role of the US as issuer of the global reserve currency.

Overall, the US robustness check confirms that while the broad transmission patterns are qualitatively consistent with the Indian case, the United States exhibits a distinct configuration of shock

contributions — one characterized by lower volatility, deeper markets, and stronger anchoring of the yield curve to short-term policy expectations.

A Appendix Figures

	Mean	StdDev	Min	Max	Skew	Kurtosis	N
IND3M_chg	0.0010	0.1618	-1.1000	2.2000	4.0485	71.8644	569.0000
IND2Y_chg	0.0024	0.1196	-0.5120	0.9560	1.5826	15.7640	569.0000
IND10Y_chg	0.0021	0.1158	-0.6200	0.7120	0.0619	12.0878	569.0000
US3M_chg	0.0025	0.0319	-0.1400	0.1900	0.2782	7.7044	569.0000
US2Y_chg	0.0015	0.0674	-0.2500	0.4000	0.5131	6.0742	569.0000
US10Y_chg	-0.0009	0.1138	-0.3400	0.3800	0.4836	3.7553	569.0000
FX_ret	0.0007	0.0468	-0.1510	0.2158	0.2406	6.5306	569.0000
NIFTY_ret	0.0026	0.0238	-0.1234	0.1436	0.4671	8.8104	569.0000
SPX_ret	0.0023	0.0211	-0.0746	0.1016	-0.3394	5.3589	569.0000

Figure 1: Descriptive Statistics

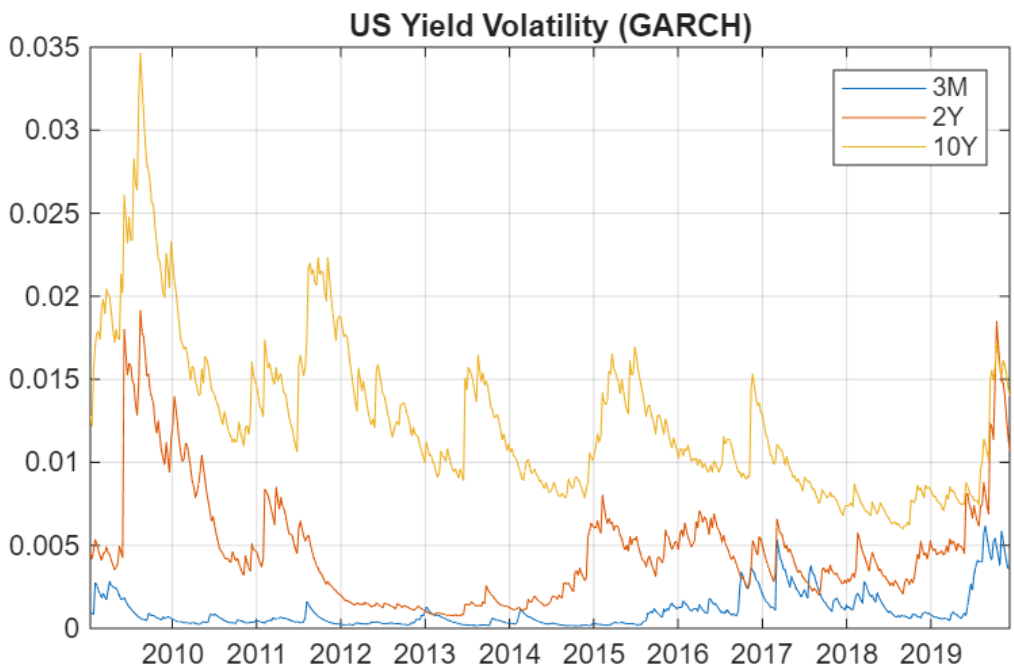

Figure A.15: U.S. Yield Volatility (GARCH)

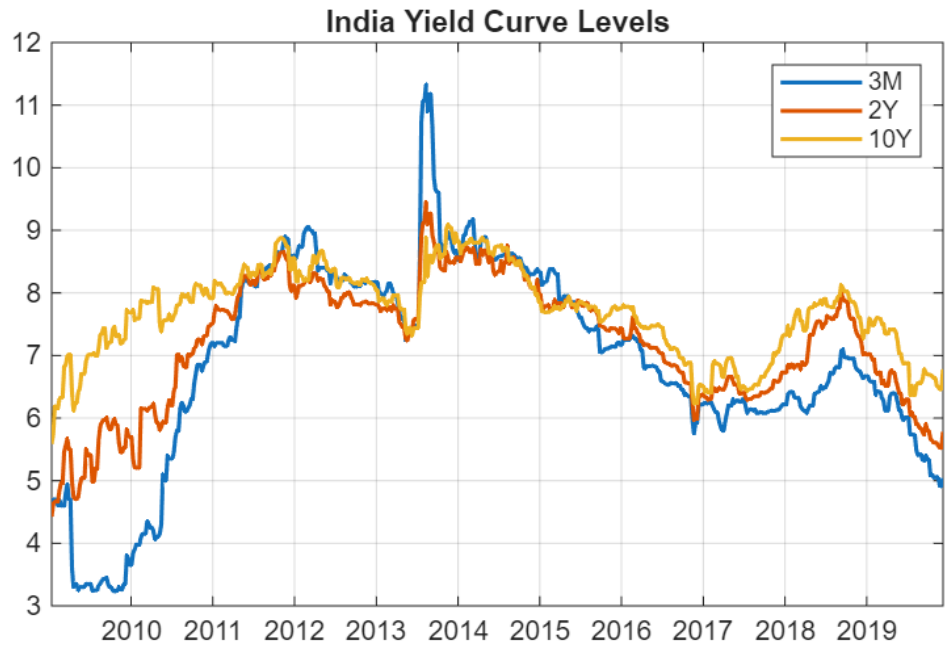

Figure A.1: India yield curve levels (3M, 2Y, 10Y).

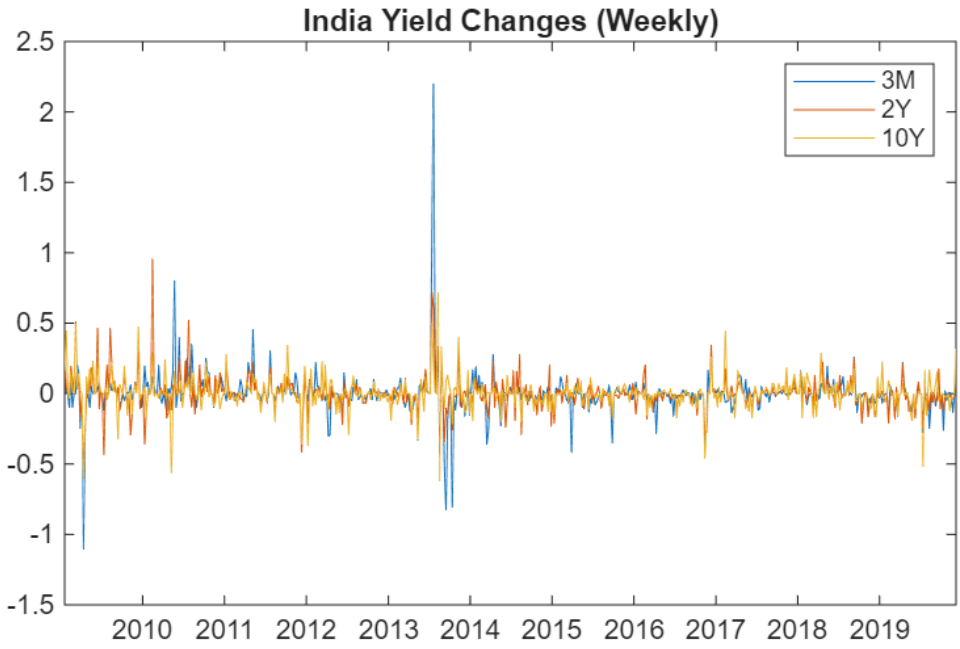

Figure A.2: Weekly changes in India yields (3M, 2Y, 10Y).

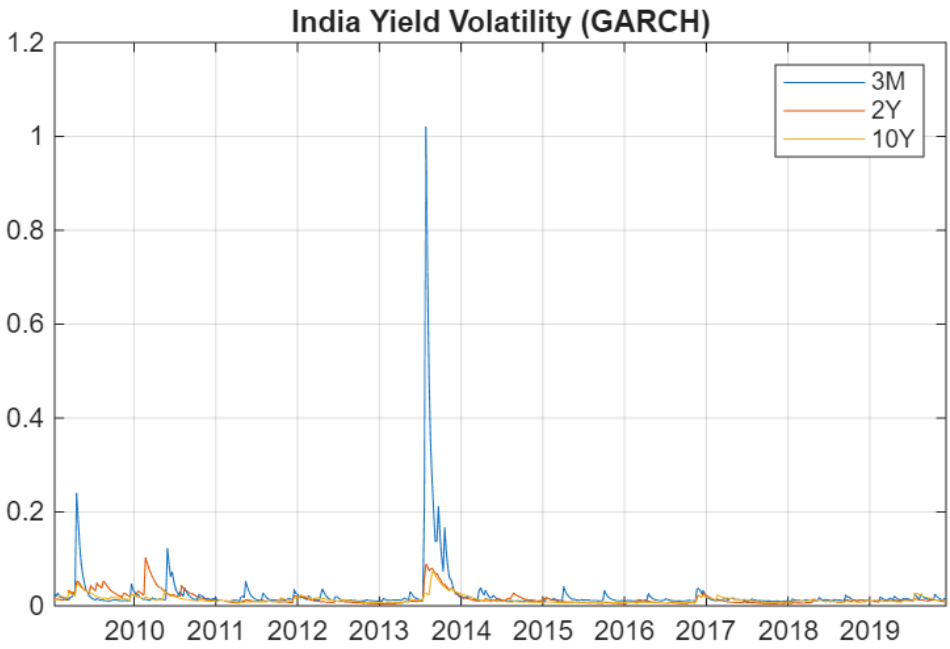

Figure A.3: Conditional volatility from GARCH(1,1) for Indian yields.

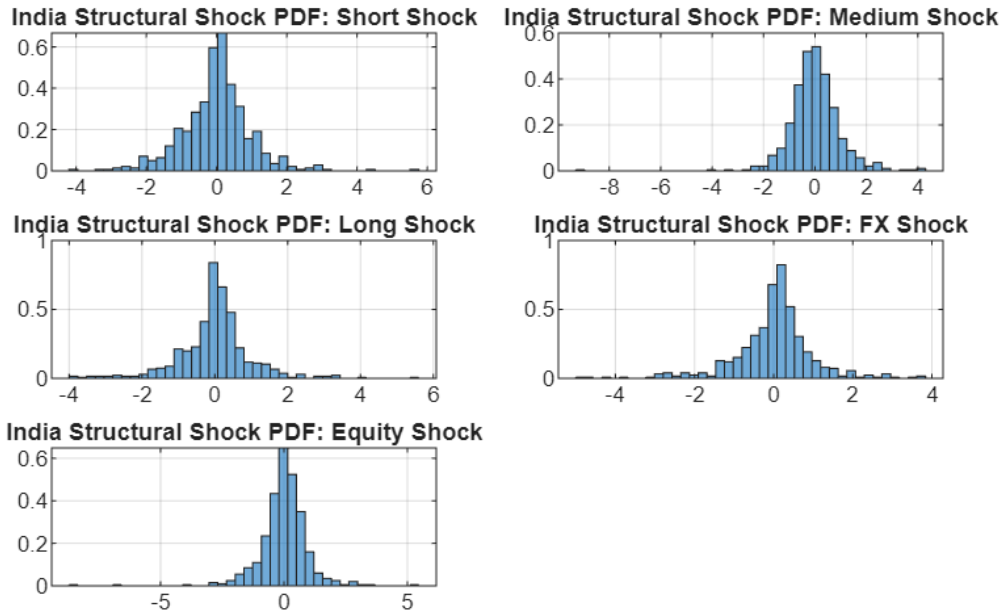

Figure A.4: Empirical distributions of structural shocks: short, medium, long, FX, equity.

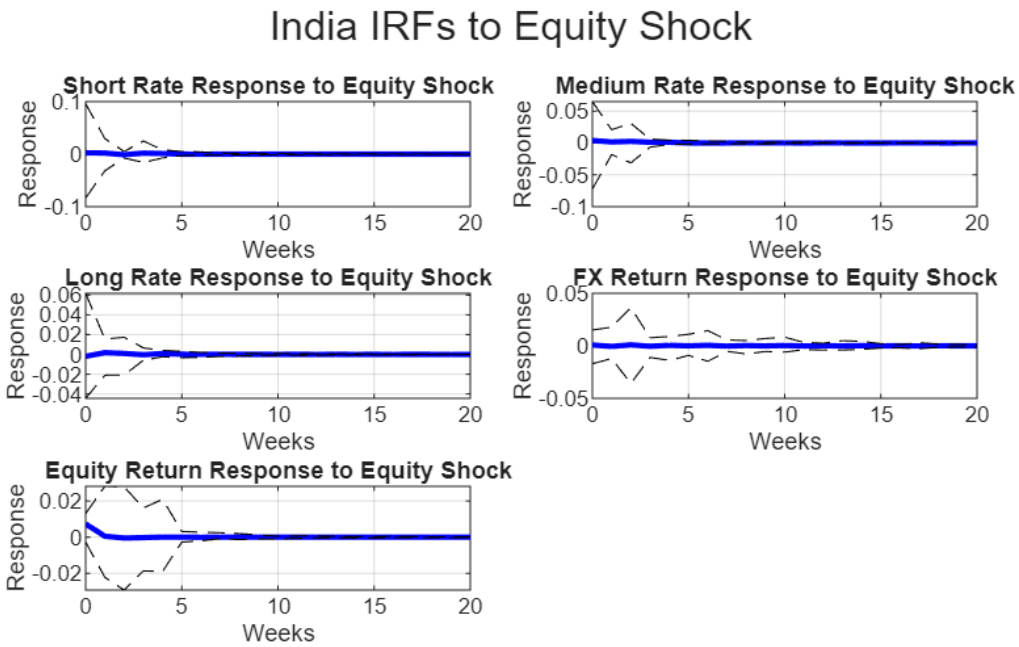

Figure A.5: Impulse responses to a short-rate shock (India).

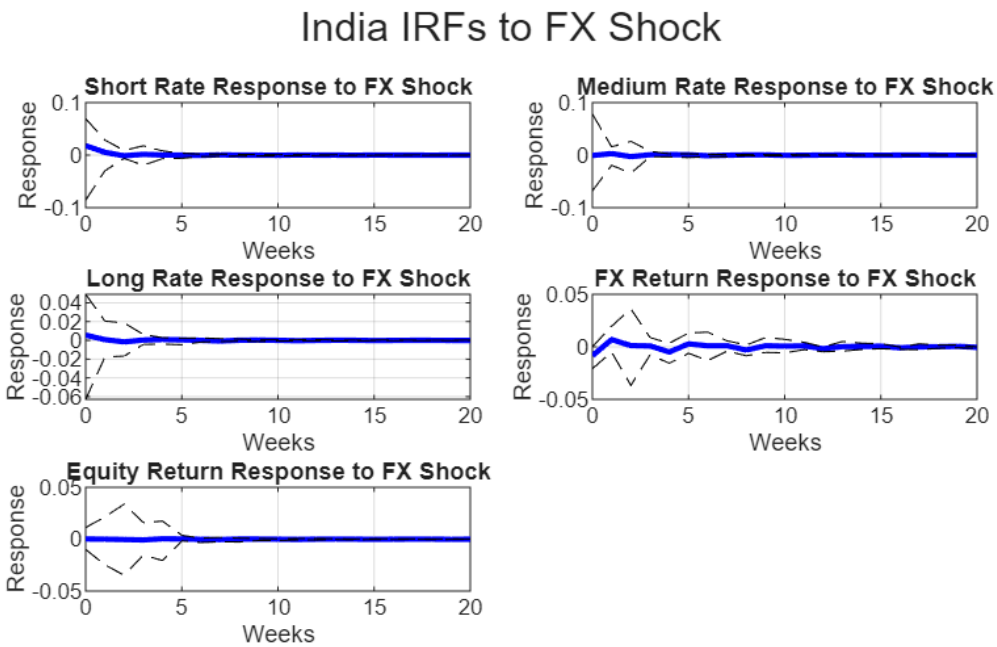

Figure A.6: Impulse responses to a medium-rate shock (India).

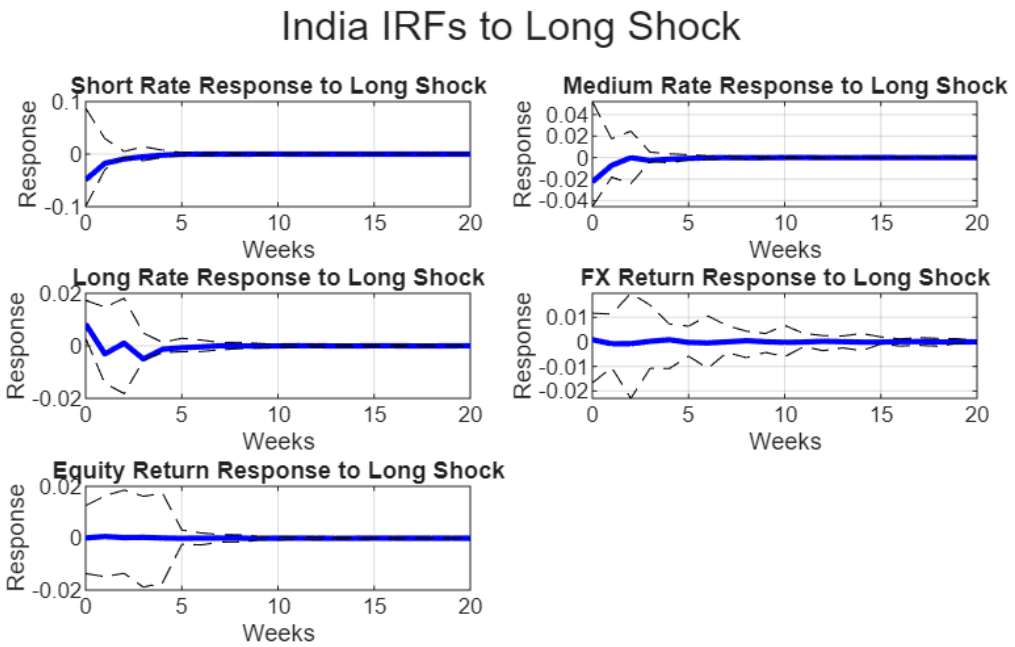

Figure A.7: Impulse responses to a long-rate shock (India).

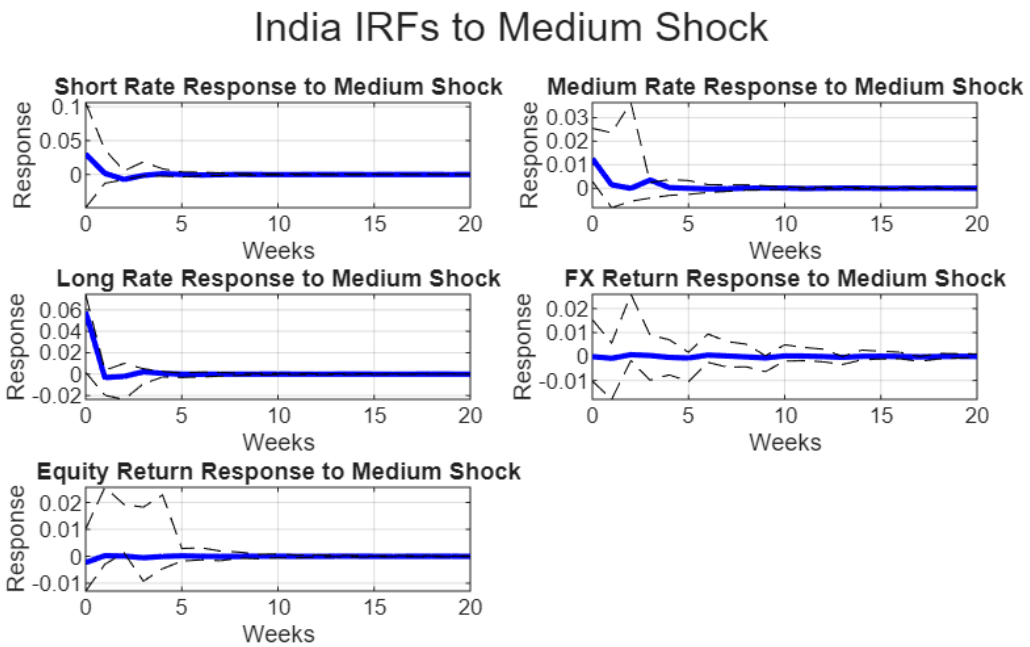

Figure A.8: Impulse responses to an FX shock (India).

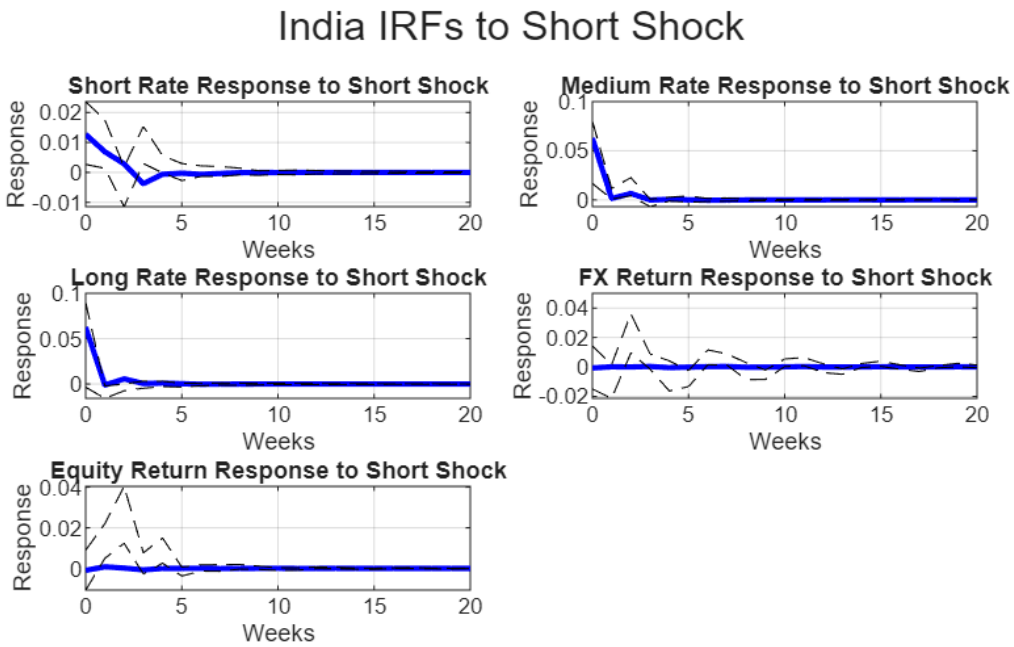

Figure A.9: Impulse responses to an equity shock (India).

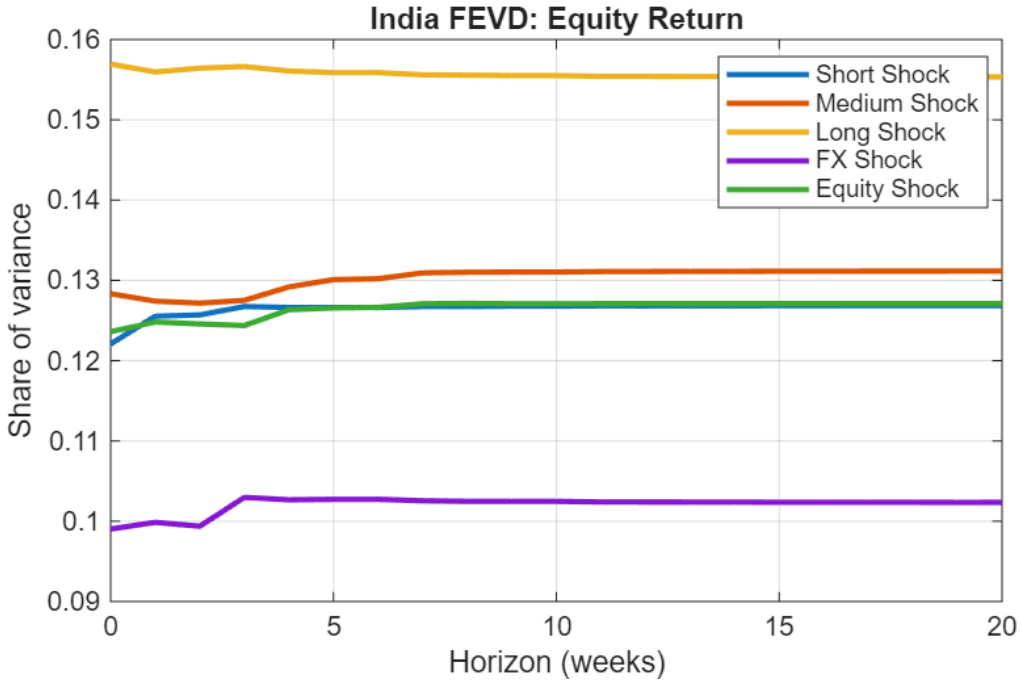

Figure A.10: FEVD: contribution of each structural shock to equity-return variance (India).

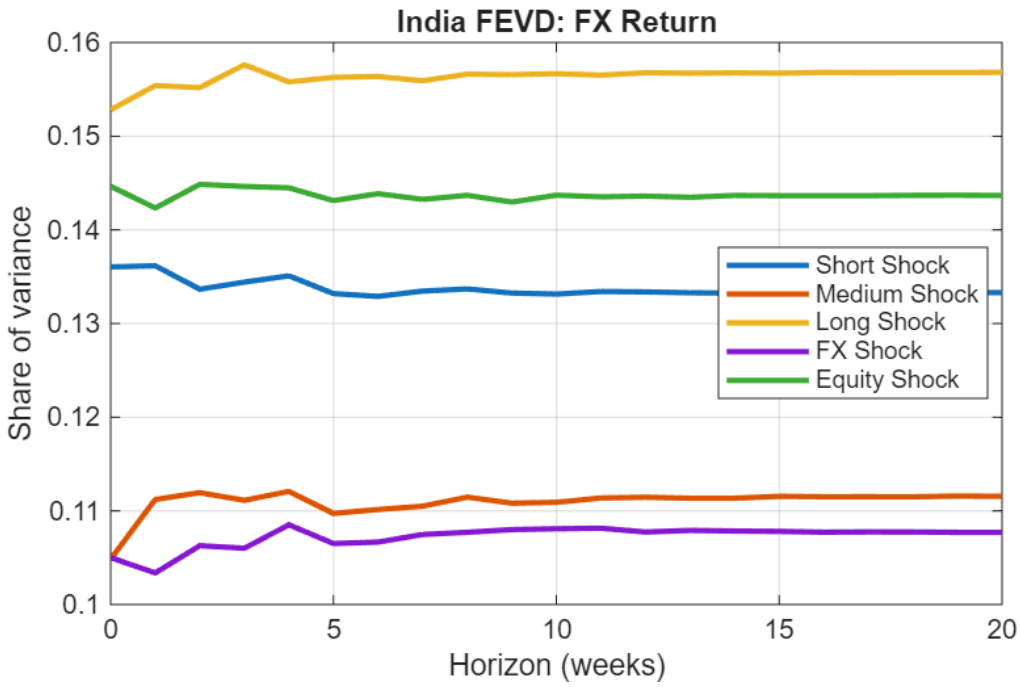

Figure A.11: FEVD: contribution of each structural shock to FX-return variance (India).

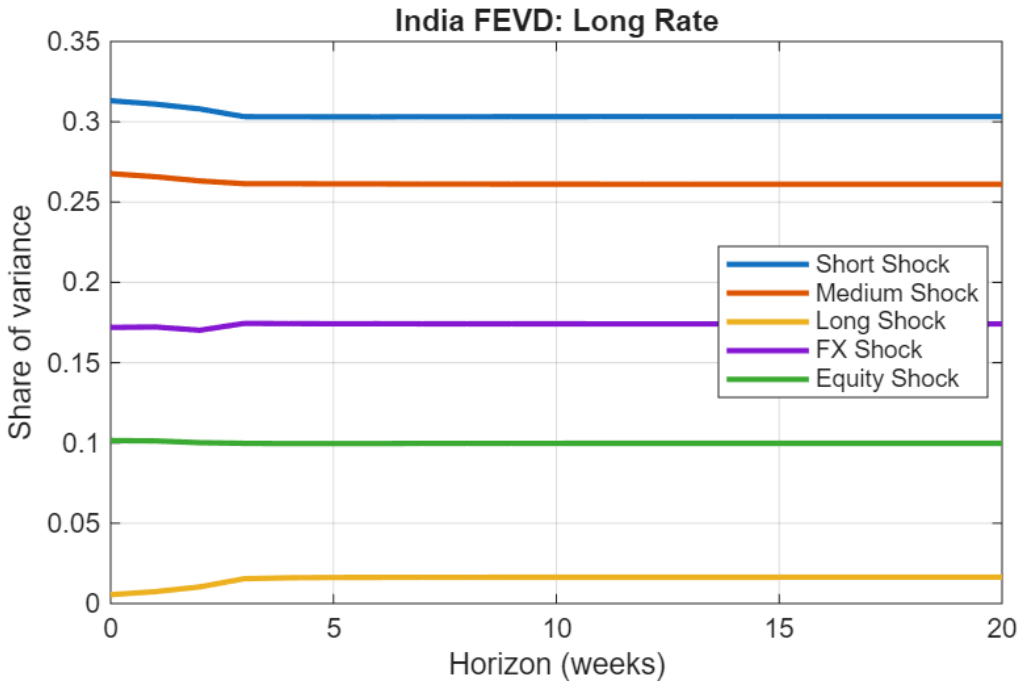

Figure A.12: FEVD: contribution of each structural shock to long-rate variance (India).

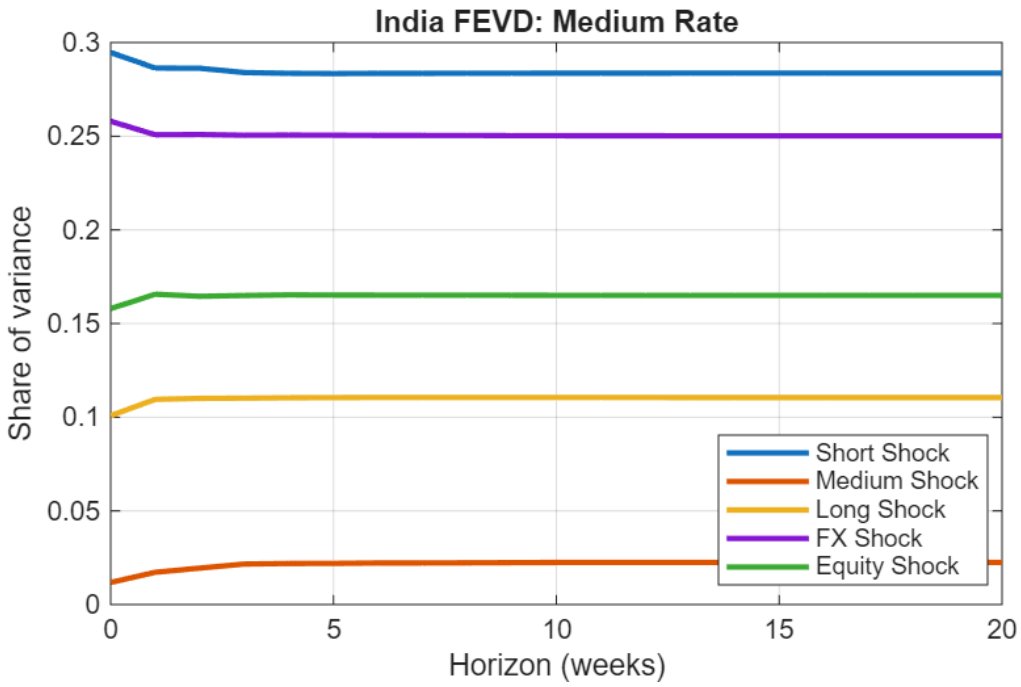

Figure A.13: FEVD: contribution of each structural shock to medium-rate variance (India).

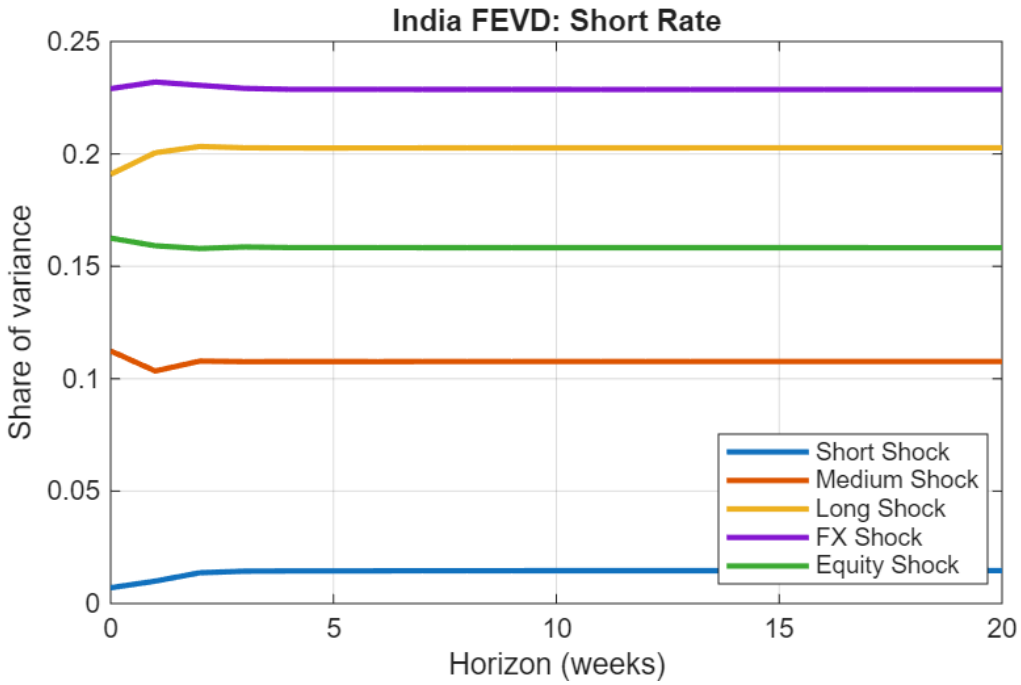

Figure A.14: FEVD: contribution of each structural shock to short-rate variance (India).

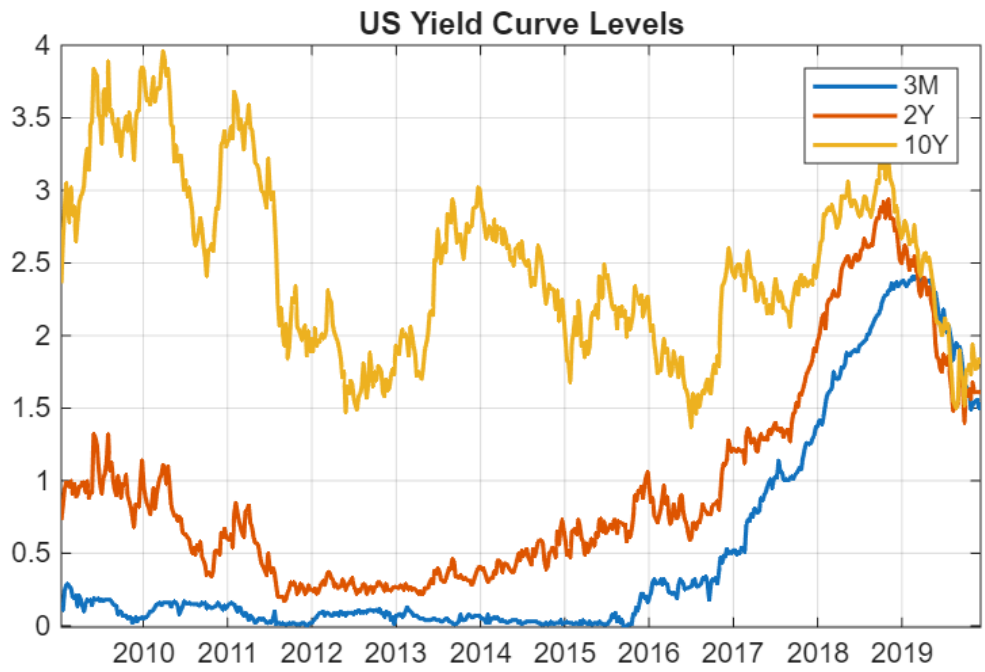

Figure A.16: U.S. Yield Curve Levels

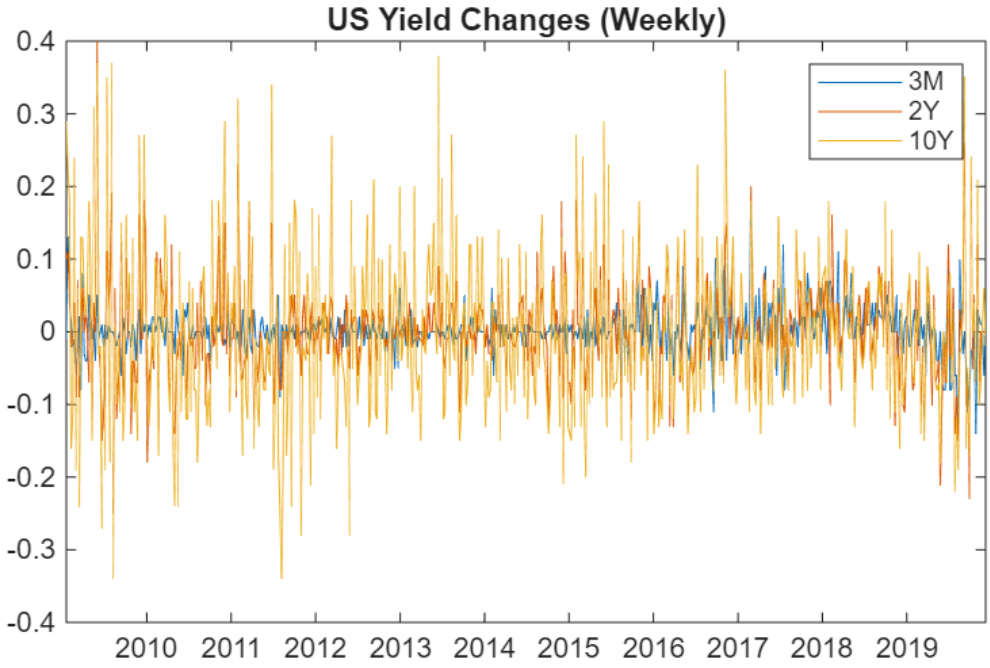

Figure A.17: U.S. Weekly Yield Changes

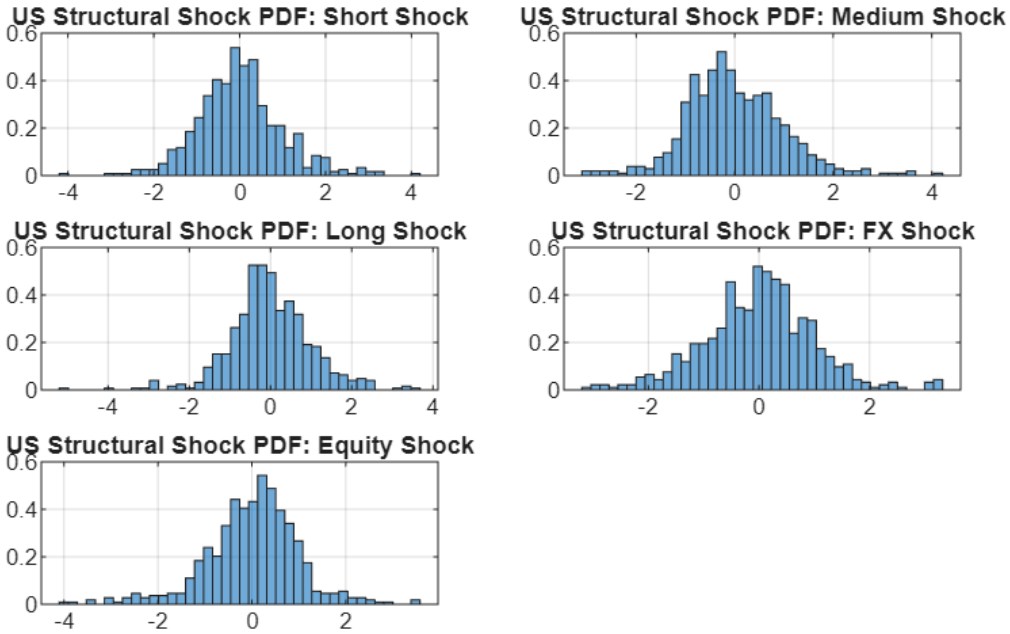

Figure A.18: U.S. Structural Shock Distributions: Short, Medium, Long, FX, and Equity Shocks

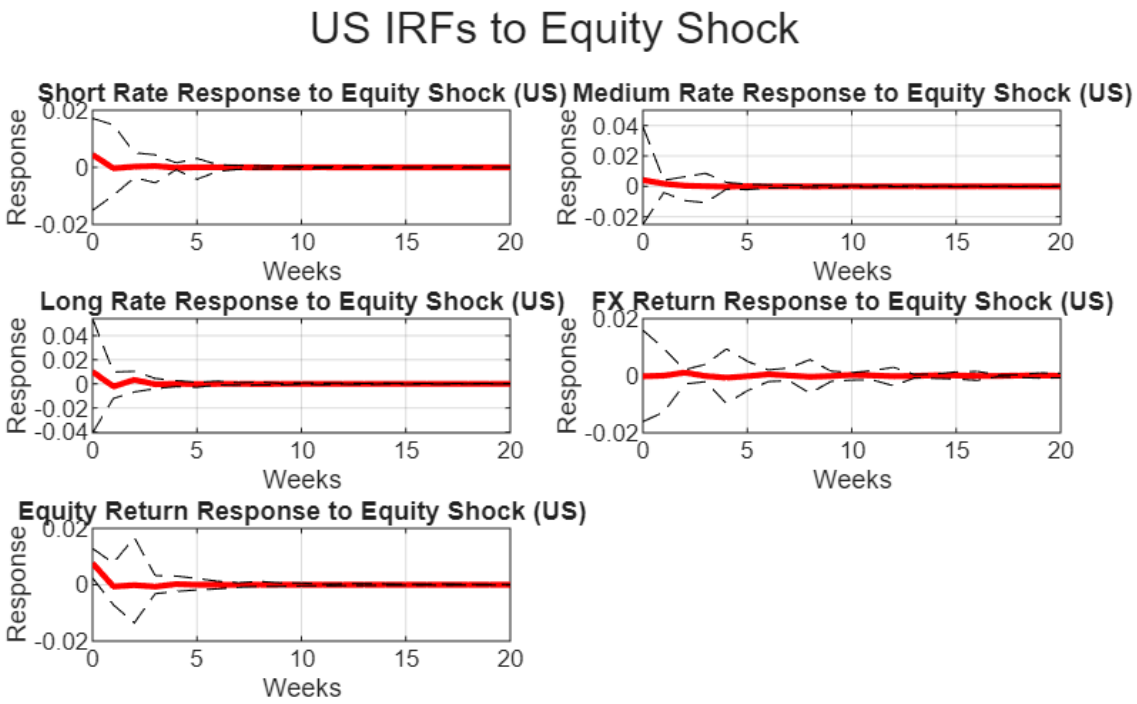

Figure A.19: A.19: US IRFs to Equity Shock

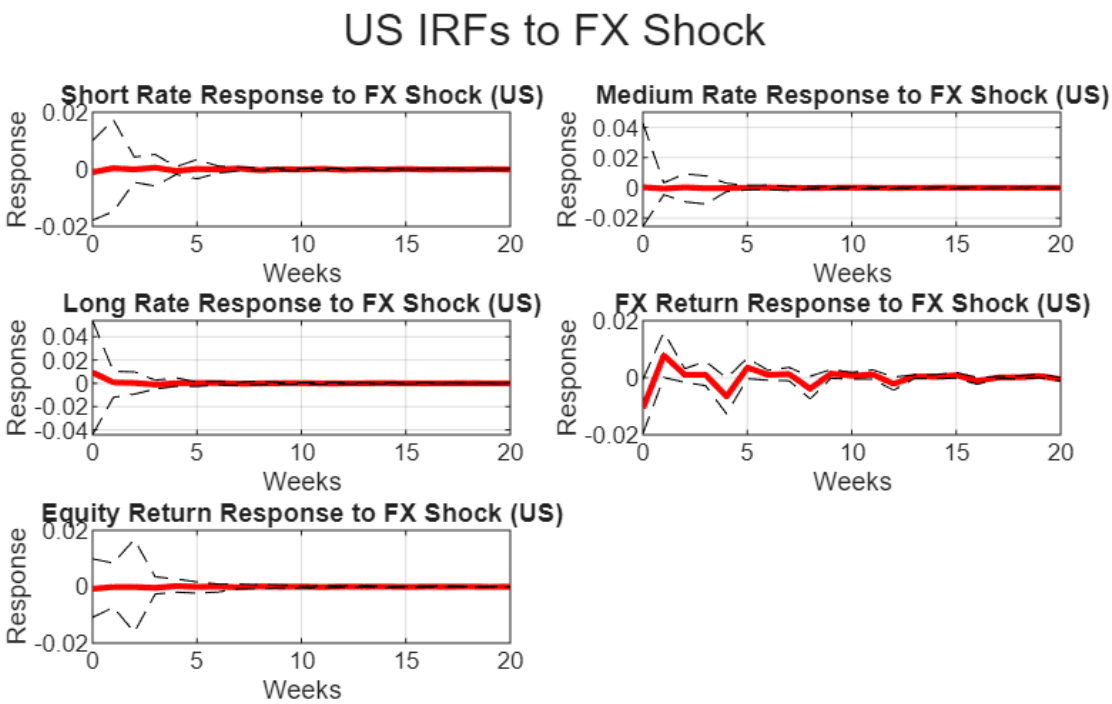

Figure A.20: A.20: US IRFs to FX Shock

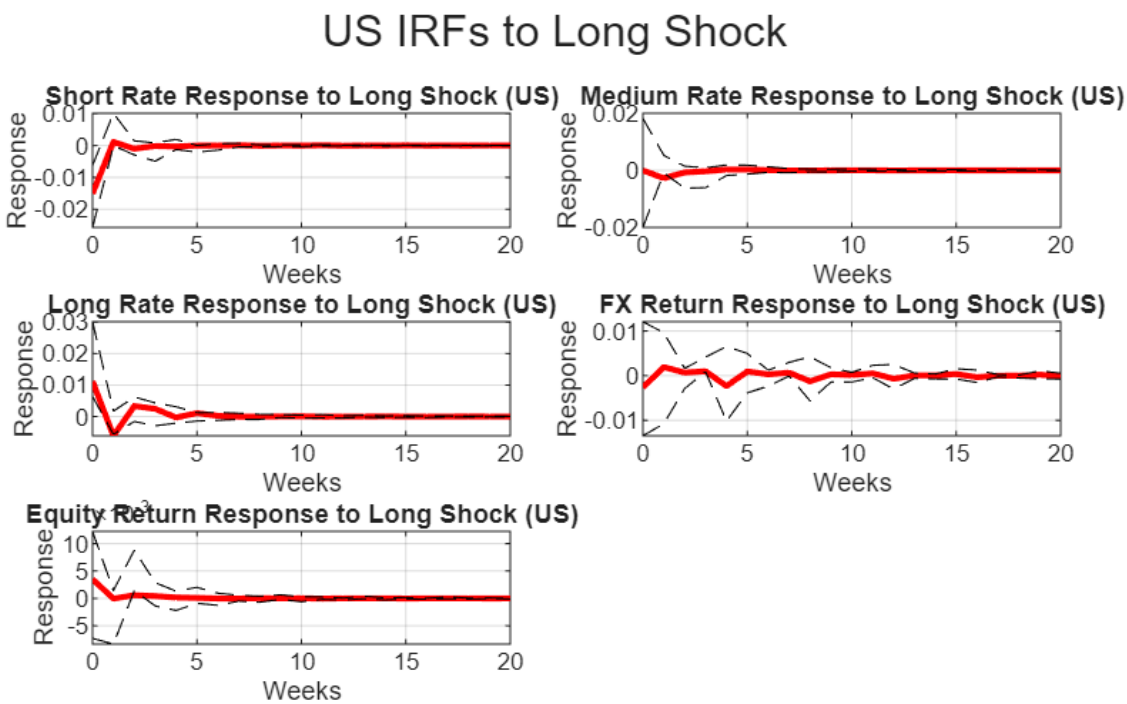

Figure A.21: A.21: US IRFs to Long-Rate Shock

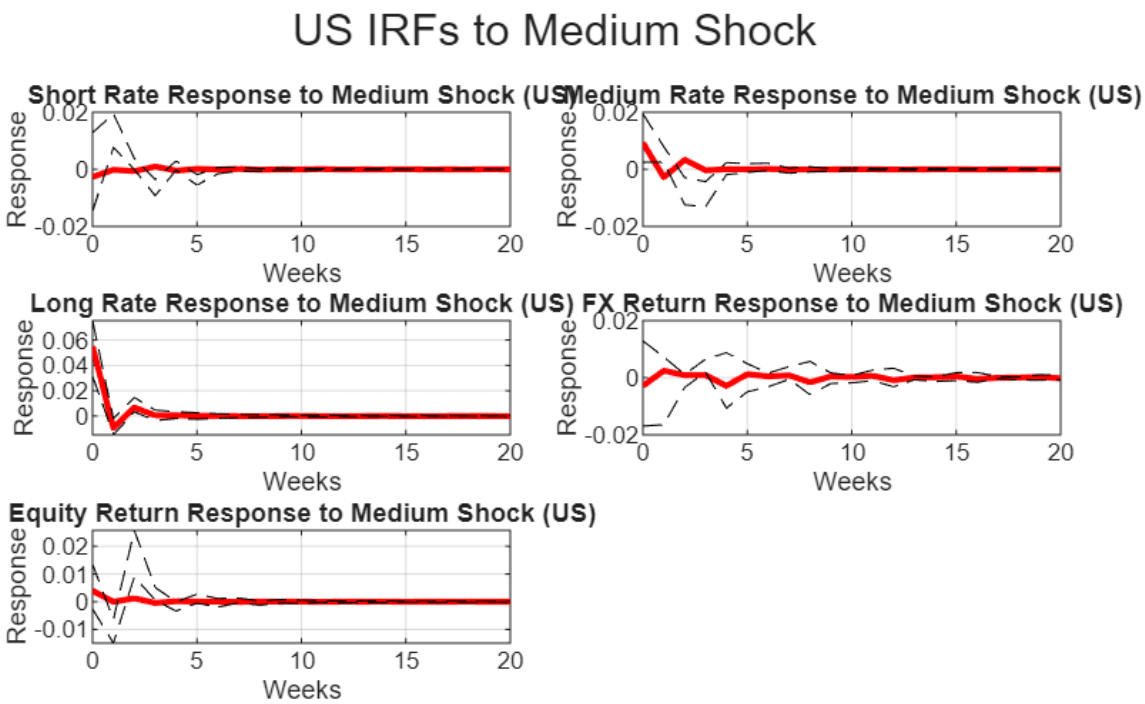

Figure A.22: A.22: US IRFs to Medium-Rate Shock

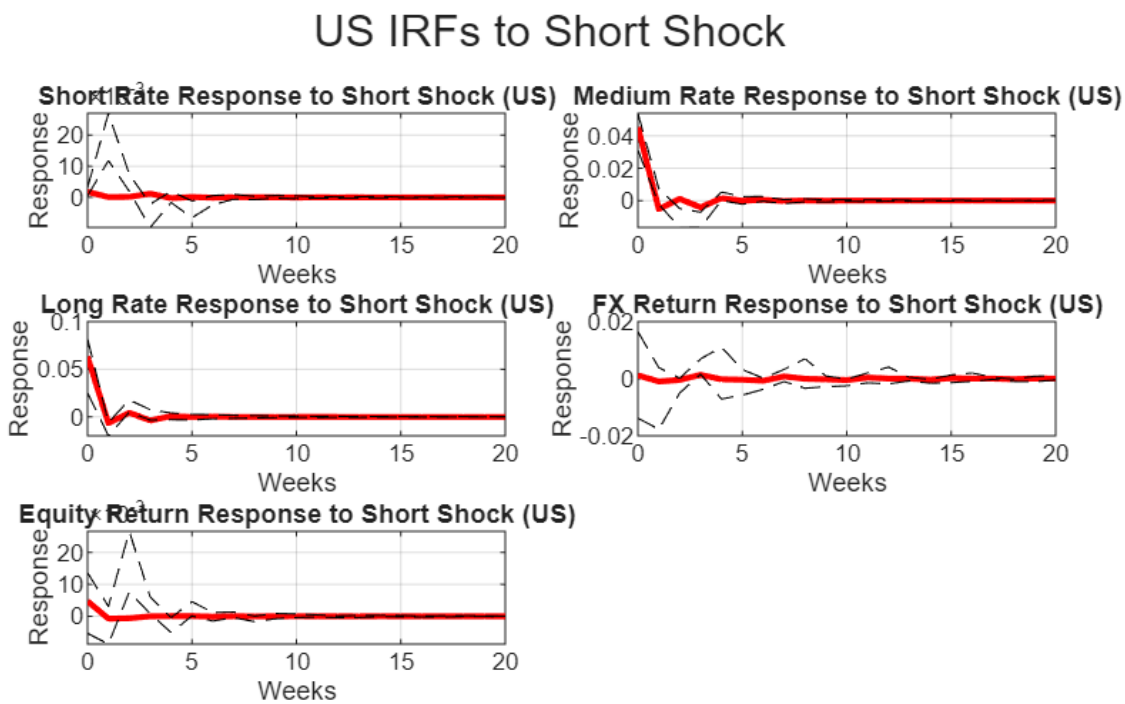

Figure A.23: A.23: US IRFs to Short-Rate Shock

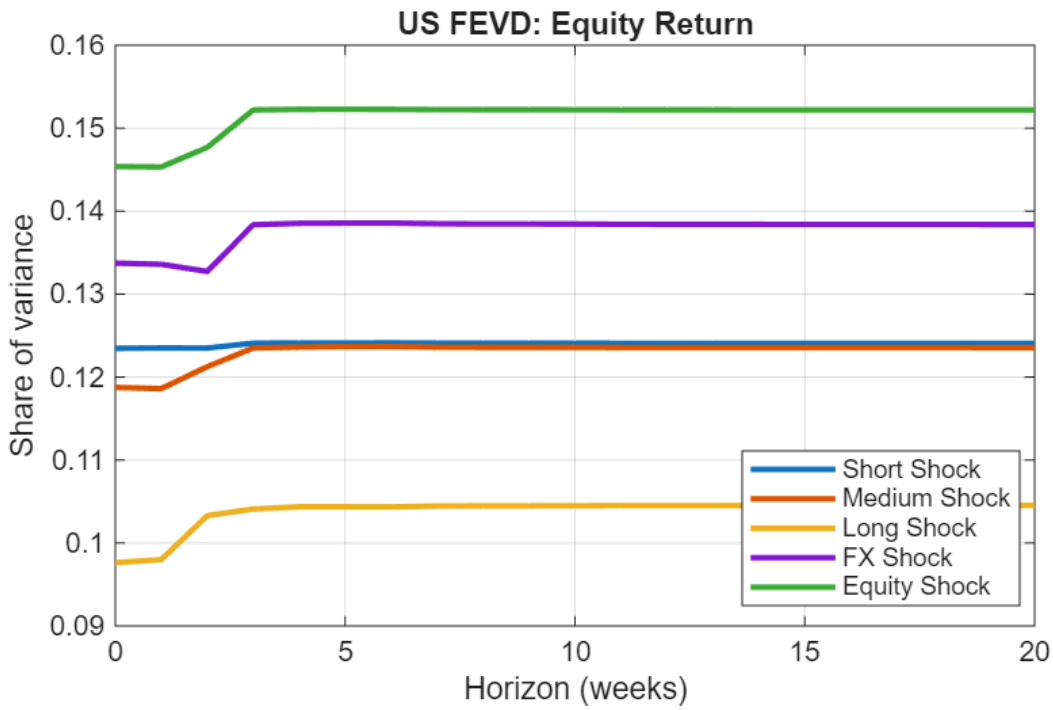

Figure A.24: A.24: US FEVD — Equity Return

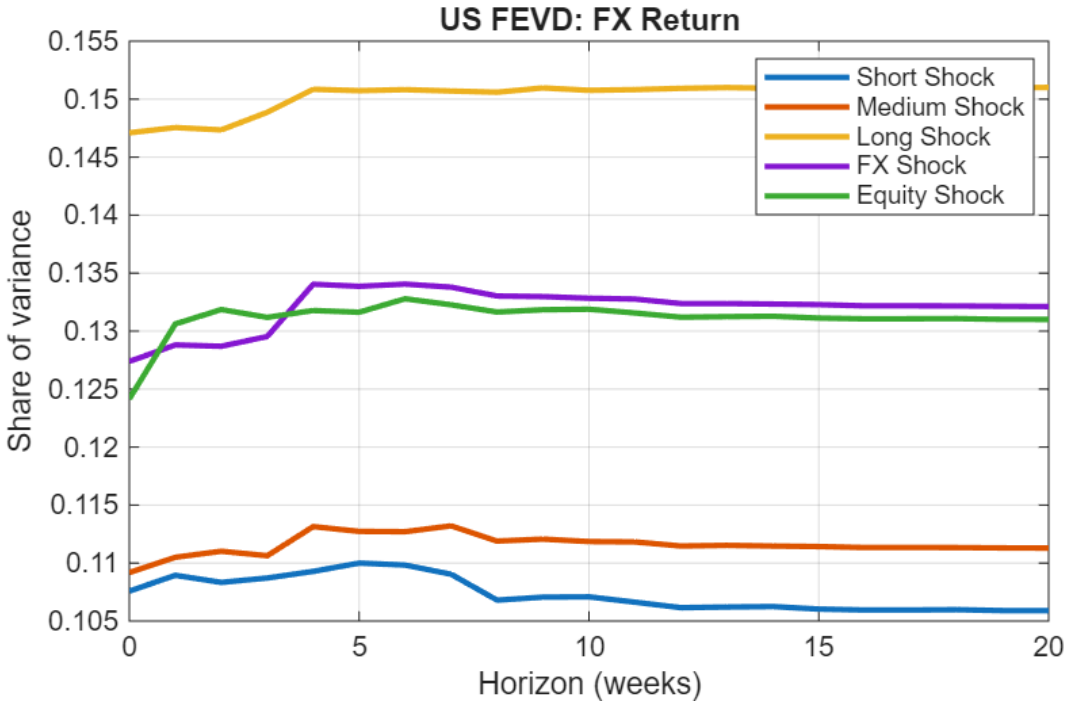

Figure A.25: A.25: US FEVD — FX Return

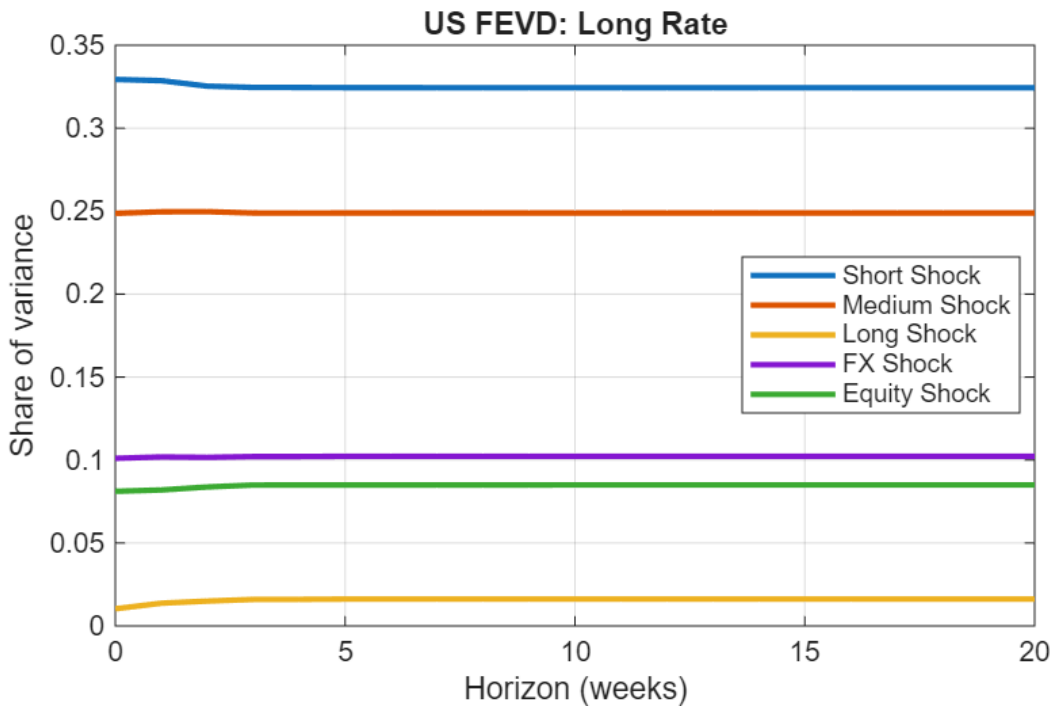

Figure A.26: A.26: US FEVD — Long-Term Yield

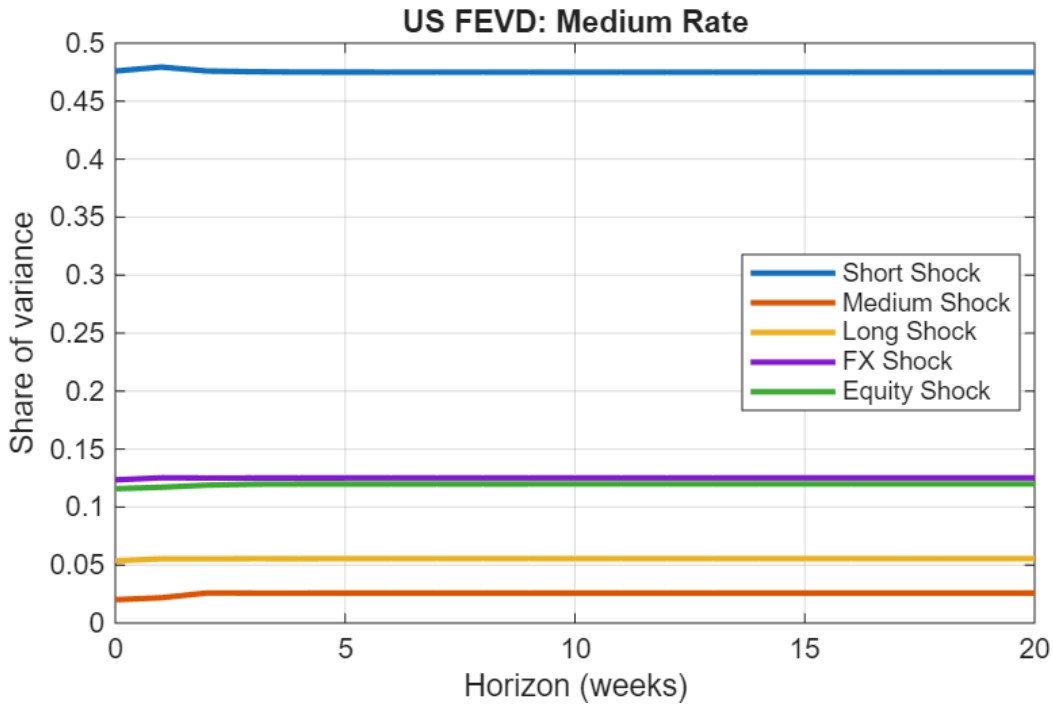

Figure A.27: A.27: US FEVD — Medium-Term Yield

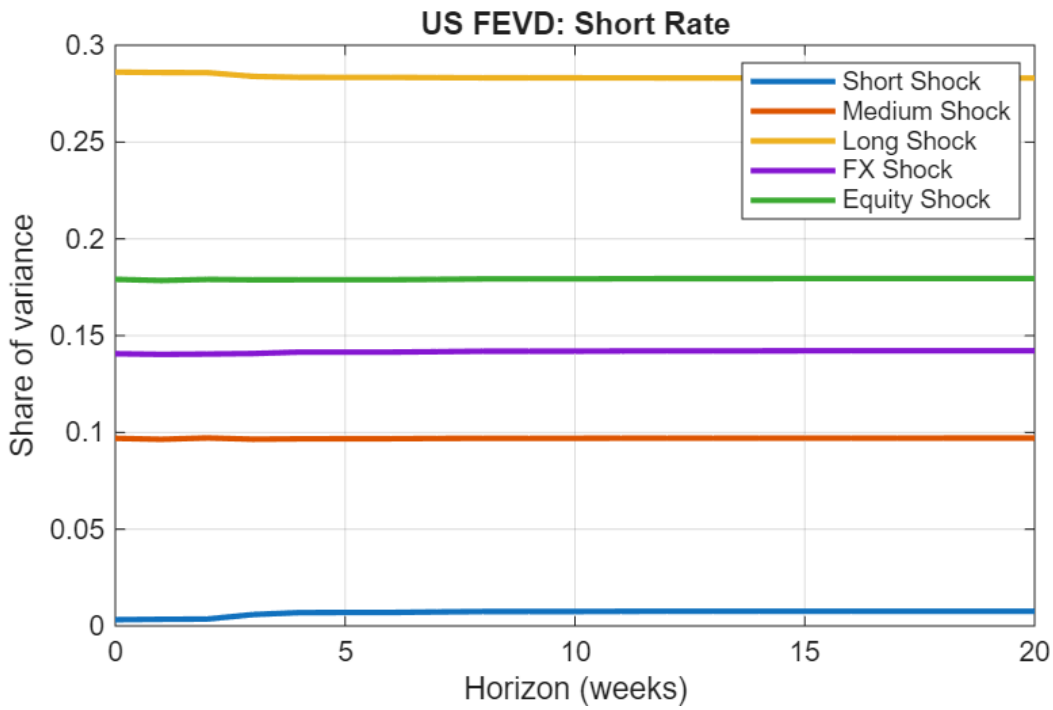

Figure A.28: A.28: US FEVD — Short-Term Yield